

ZEWEI DENG

Ph.D. Student in Industrial and Systems Engineering | University of Minnesota Twin Cities

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U.S. Work Authorization · No Employer Sponsorship or CPT Required

PROFILE

Develop statistical, coding-theoretic, and optimization methods for trustworthy AI, with research on LLM watermarking and provenance, distributionally robust inference, uncertainty-aware sequential modeling, and temporal point processes.

EDUCATION

University of Minnesota Twin Cities

Ph.D. in Industrial and Systems Engineering; Advisor: Prof. Liyan Xie

2025–Present
Minneapolis, MN

University of Chicago

M.S. in Applied Data Science

2023–2024
Chicago, IL

University of Illinois Urbana-Champaign

B.S. in Statistics; Minors in Computer Science and Economics; GPA: 3.9/4.0

2019–2023
Urbana-Champaign, IL

RESEARCH & PROFESSIONAL EXPERIENCE

Graduate Research Assistant, University of Minnesota Twin Cities

Advisor: Prof. Liyan Xie

Summer 2026–Present
Minneapolis, MN

- Develop statistical and coding-theoretic methods for LLM watermark provenance, edit detection, local integrity verification, and watermarkability.
- Led an ECC watermark project accepted to EMNLP 2026 Findings, spanning method design, implementation, experiments, and manuscript preparation.
- Develop uncertainty-aware sequential models and distributionally robust inference for delayed point processes under reporting delays and model misspecification.

Research Assistant, Chicago Booth School of Business

Advisor: Prof. Dacheng Xiu

2024
Chicago, IL

- Built financial-text forecasting pipelines using BERT, LLaMA, and T5 representations with regression models for return and debt-related prediction.

Machine Learning Capstone, Alliant Credit Union / UChicago DSI

Client-facing machine-learning project

2024
Chicago, IL

- Contributed to model development and evaluation for a credit-union analytics application.

PUBLICATIONS & WORKING PAPERS

Detecting Post-generation Edits to Watermarked LLM Outputs via Error-Correcting Codes.

Zewei Deng, Muhammad Siddeek, Liyan Xie, Mohamed Seif, Mengdi Wang, H. Vincent Poor, and Andrea Goldsmith.
EMNLP Findings, 2026.

Anchor-ECC: Local Integrity Checking for Watermarked LLM Outputs via Error-Correcting Codes.

Zewei Deng, Muhammad Siddeek, Liyan Xie, Mohamed Seif, Mengdi Wang, H. Vincent Poor, and Andrea Goldsmith.
arXiv:2609.38722, 2026; planned submission to Transactions on Machine Learning Research (TMLR). [Paper]

Distributionally Robust Inference for Delayed Point Processes.

Zewei Deng, Shuyi Chen, Shixiang Zhu, and Liyan Xie.

Working paper in preparation for submission to the Annals of Applied Statistics (AOAS); workshop version forthcoming at the INFORMS Data Mining and Decision Analytics Workshop, 2026.

Watermarkability: Measuring and Improving Language-Model Compatibility with Logit-Based Watermarks.

Zewei Deng, Mohamed Seif, and Liyan Xie.

Preprint, 2026.

HawkesLLM: Semantic Uncertainty Propagation in Agentic Text Simulation.

Zewei Deng, Tinghan Ye, and Liyan Xie.

ICML Workshop on Statistical Frameworks for Uncertainty in Agentic Systems, 2026. [Paper]

Learning to Test: Physics-Informed Representation for Dynamical Instability Detection.

Minxing Zheng, **Zewei Deng**, Liyan Xie, and Shixiang Zhu.

arXiv:2604.10967, 2026. [Paper]

TEACHING & HONORS

Teaching: Teaching Assistant, University of Minnesota Twin Cities, Sep 2025–May 2026

Honors: Highest Distinction in Statistics; Dean's List (2020–2023); COMAP MCM Honorable Mention (2021)